

SURGE-PROMPT: SURROGATE-GUIDED, ROBUST, AND MODULAR PROMPT OPTIMIZATION FOR BLACK-BOX LLMs

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ABSTRACT

We address practical and scientific gaps in black-box prompt optimization for large language models, where existing approaches are token-expensive, brittle to phrasing and version drift, hard to audit, and lack calibrated reliability. We introduce SURGE-Prompt, a framework that searches over a typed, modular Prompt Intermediate Representation with a validator; learns an uncertainty-calibrated surrogate over fragments to predict task success, format and safety risks, and cost; and acquires candidates via constrained, saliency-guided edits with distributionally robust selection across paraphrases and model versions. SURGE provides conformal lower confidence bounds for budgeted reliability, early stopping to reduce waste, and an optional instance-adaptive router to trade cost for quality. It integrates as a drop-in `propose()/learn()` module in OPRO/EvoPrompt/ZOPO-style loops, adding pre-call validation and drift recalibration. We verify SURGE in three studies: (1) budgeted effectiveness and calibration against OPRO-, EvoPrompt-, and ZOPO-like baselines; (2) robustness under paraphrase and model-version shifts with few-shot recalibration and a router; and (3) ablations on calibration, the validator, saliency-guided edits, and causal A/B attribution. Across a synthetic, governance-aware evaluation, SURGE reaches comparable or better quality with fewer calls, lowers schema and safety violations, attains near-target empirical coverage, improves worst-case robustness, and reduces tokens via routing.

1 INTRODUCTION

Prompt optimization is now a primary tool for adapting large language models (LLMs) when only black-box access is available. Yet mainstream optimizers

LLM-as-optimizer loops, evolutionary mutations, and pool-based local search

are often token-inefficient, brittle to phrasing and provider updates, hard to audit, and provide no calibrated reliability on success, schema adherence, or safety. In enterprise LMaaS settings, teams demand predictable budgets, constraint safety, explainable edits, and auditable reliability.

Prior work illuminates parts of this space but leaves key gaps. LLM-as-optimizer frameworks such as OPRO iteratively generate solutions with textual feedback but do not furnish calibrated uncertainty, validator-backed safety, or robustness to paraphrase and version drift Yang et al. (2023). Evolutionary methods like EvoPrompt and Promptbreeder automate mutation and selection, yet operate on free-form strings without typed, immutable fragments or risk calibration [guo2023connecting, fernando2023promptbreeder]. Pool-based local optimization (e.g., ZOPO) improves query efficiency via kernelized surrogates, but typically cannot synthesize quality prompts and lack calibrated reliability Huet al. (2024). RL-based discrete optimization (RLPrompt) explores prompt spaces but generally forgoes validator-backed safety and conformal guarantees Deng et al. (2022). Continuous or soft prompt tuning in LMaaS requires internal access and compromises interpretability and governance Sun et al. (2022). In-based discrete prompt discovery focuses on white-box settings Wen et al. (2023), and prompt adaptation for text-to-image is modality-specific and lacks calibrated reliability for NLP tasks Hao et al. (2022). Related advances for VLM prompts but without calibrated uncertainty or validator-backed governance Liu et al. (2023).

We propose SURGE-Prompt, a surrogate-guided, robust, and modular optimization framework for black-box LLMs. SURGE operates over a typed Prompt Intermediate Representation (PIR) with a validator that enforces immutability, schema, and safety constraints before any API call. A calibrated deep-ensemble surrogate over fragments predicts success, violation risks, and cost, and provides conformal lower confidence bounds (LCBs). Acquisition uses saliency-guided, constrained span edits and optimizes distributionally robust objectives across paraphrases and plausible model-version shifts. Early stopping prunes low-promise candidates with Wilson bounds. A small contextual bandit router toggles optional fragments (e.g., reasoning, self-check) per instance to trade tokens for accuracy while respecting constraints. A few-shot drift adapter restores coverage on new model versions through temperature and conformal recalibration with residual correction.

1.1 CONTRIBUTIONS

- **Typed, validated Prompt IR:** Interpretable, constraint-safe editing under audit, with fragment banks for reuse.
- **Calibrated surrogate with conformal bounds:** A deep-ensemble over fragment graphs using temperature scaling and split conformal prediction to produce per-candidate LCBs for success and UCBs for violations.
- **Saliency-guided constrained edits:** Mask-and-infill edits guided by attribution, plus micro-batch causal A/B toggles to localize high-leverage fragments and update the surrogate online.
- **Distributionally robust acquisition and early stopping:** Robust objectives across paraphrase and model panels (mean minus alpha times standard deviation or CVaR), diversification, and Wilson-bound pruning.
- **Active learning, router, and recalibration:** Conformal guardrails for exploration, an instance-adaptive router for cost-quality control, and few-shot drift recalibration with residual correction.
- **Drop-in integration API:** A minimal propose()/learn() interface that plugs into OPRO/EvoPrompt/ZOPO-style loops [yang2023_{large}, guo2023_{connecting}, hu2024_{localized}].

We validate SURGE in three experiments. First, we study budgeted effectiveness, efficiency, and calibration against OPRO-, EvoPrompt-, and ZOPO-like baselines. Second, we assess robustness under paraphrases and model-version shifts and test few-shot recalibration and routing. Third, we analyze surrogate calibration, saliency-guided edits, and causal interpretability with ablations. Across a synthetic, governance-aware setup, SURGE achieves comparable or better quality with fewer calls, near-target coverage, fewer violations, improved worst-case robustness, and token savings. We conclude with limitations and future work on real-world drift and multimodal extensions.

2 RELATED WORK

2.1 LLM-AS-OPTIMIZER

OPRO frames optimization as iterative solution generation conditioned on prior solutions and scores, showing strong gains for prompt optimization but without calibrated uncertainty, validator-backed governance, or robustness objectives Yang et al. (2023). SURGE complements OPRO by replacing free-form mutation with validated fragment edits and by ranking with calibrated LCBs.

2.2 EVOLUTIONARY SEARCH

EvoPrompt connects LLM mutations with evolutionary operators to improve diverse tasks efficiently Guo et al. (2023). Promptbreeder co-evolves task and mutation prompts and improves reasoning performance Fernando et al. (2023). Both optimize free-form strings without typed fragments, hard immutability, or conformal calibration. SURGE introduces a typed PIR and a validator, enabling safety and schema guarantees, and adds calibrated uncertainty.

2.3 POOL-BASED LOCAL OPTIMIZATION

ZOPO argues high-performing local optima are prevalent and uses a kernelized Gaussian-process proxy over prompt embeddings to search locally Hu et al. (2024). While efficient, local search often relies on a fixed pool and lacks mechanisms to synthesize constraint-safe novel prompts or to provide calibrated risk. SURGE generates new, validated candidates and learns a calibrated surrogate across tasks and model versions.

2.4 REINFORCEMENT LEARNING FOR DISCRETE PROMPTS

RLPrompt formulates prompt generation with a policy optimized by reward, but generally omits validator-backed safety and uncertainty calibration Deng et al. (2022). SURGEs active learning and router share RL spirit yet operate under hard constraints and conformal guardrails.

2.5 INSTRUCTION DISCOVERY AND TOOLING

Automatic Prompt Engineer creates instruction candidates via an LLM and selects by development performance Zhou et al. (2022). OpenPrompt offers modular prompting components Ding et al. (2021). These contributions inform components of our system but do not provide distributional robustness, conformal guarantees, or typed, immutable safety/schema fragments.

2.6 CONTINUOUS OR WHITE-BOX APPROACHES

Black-Box Tuning for LMaaS optimizes soft prompts with derivative-free methods but requires internal access and reduces interpretability Sun et al. (2022). Gradient-based discrete prompt discovery likewise assumes white-box signals Wen et al. (2023). SURGE targets strictly black-box APIs while retaining interpretability and governance through fragments and pre-call validation.

2.7 MULTIMODAL EXTENSIONS

LLMs have been used as black-box optimizers for VLMs, improving prompts with conversational feedback Liu et al. (2023), and prompt adaptation has advanced text-to-image aesthetics Hao et al. (2022). These works motivate the generality of prompt optimization but lack calibrated UQ and validator-backed governance that SURGE provides for NLP and that can extend to VLMs.

In summary, prior methods either optimize free-form strings without guarantees [yang₂₀₂₃_{large}, guo₂₀₂₃_{connecting}, fernando₂₀₂₃_{promptbreeder}], focus on local pools without synthesis or calibration in interfaceto existing loops.

3 BACKGROUND

3.1 PROBLEM SETTING

We consider a black-box LLM f that maps a prompt p and input x to an output y , with no gradient or logits access. For a task T with metric $M(y, x)$, the goal is to find a prompt p^* that maximizes expected performance under hard constraints on output schema and safety, subject to a finite budget of model calls. The environment is nonstationary: paraphrases of x and model version changes induce distribution shifts that can degrade performance.

3.2 PROMPT INTERMEDIATE REPRESENTATION

SURGE represents prompts as sequences of typed fragments: reasoning trigger, step plan, tool/verifier hints, output schema, tone/style, safety guardrails, and brevity/cost policy. Fragments carry metadata (max tokens, reserved markers, immutability, schema references, provenance). A validator enforces: ordering (e.g., safety and schema precede reasoning), JSON Schema conformance, regex-based markers, per-fragment and global length limits, immutability of safety/schema, and guardrail monotonicity. This modularization turns free-form string search into constrained, interpretable editing, enabling governance and reuse.

3.3 SURROGATE MODELING AND CALIBRATION

We learn a predictive surrogate over fragmentized prompts that ingests: the fragment graph with types and order; a task descriptor; prompt statistics; optional proxy scores from a small open-source model on micro-batches; a model-ID embedding encoding provider family/version/decoding; and optional input-difficulty features. A fragment-graph Transformer with a deep ensemble outputs success probability, format- and safety-violation probabilities, and a cost prediction. Temperature scaling calibrates probabilities, and split conformal prediction yields lower confidence bounds for success and upper bounds for violations, enabling budgeted reliability.

3.4 ROBUST ACQUISITION

To improve robustness to paraphrases and model drift, SURGE evaluates candidate prompts under a panel formed by input paraphrases and model-ID embeddings representing plausible version shifts. It optimizes a robust objective such as mean minus alpha times standard deviation or the 10 percent conditional value-at-risk (CVaR), subject to hard violation caps and soft cost penalties. Batch selection is diversified to avoid mode collapse, and early stopping uses Wilson bounds during dev evaluation to conserve budget.

3.5 ACTIVE LEARNING, ROUTING, AND DRIFT ADAPTATION

SURGEs learner prioritizes high-uncertainty or high-surprise candidates and enforces exploration within conformal error budgets. A small contextual bandit router toggles optional fragments per instance to trade accuracy and tokens under constraints. For model-version drift, few-shot recalibration updates temperature and conformal thresholds and adds a residual corrector without full retraining, restoring coverage on new versions.

3.6 CONTEXT WITHIN LITERATURE

This setup unifies advantages of instruction discovery and evolutionary search [zhou₂₀₂₂*large*, guo₂₀₂₃*connecting*, fernando₂₀₂₃*promptbreeder*], *localizedpool* — *basedoptimization* Huet *al.* (2024), and *RL-styleadaptivity* Denget *al.* (2022) while addressing black-box governance requirements absent in prior methods.

4 METHOD

4.1 TYPED PIR AND VALIDATOR

SURGE segments raw prompts into typed fragments and restricts edits to insert, delete, swap, move, and mask-infill at fragment or span level. `safety_guardrails` and `output_schema` fragments are immutable by default. Per-type and global length ceilings are enforced; reserved markers are checked by regex; schemas are validated via JSON Schema. A fragment bank stores reusable components with metadata and historical metrics for cross-task reuse and audit. The validator rejects any edit that violates ordering, immutability, length, markers, or schema rules, ensuring pre-call constraint safety.

4.2 UNCERTAINTY-AWARE SURROGATE OVER FRAGMENTS

Inputs comprise: (i) the fragment graph with type and order encodings; (ii) a task descriptor and exemplar summary; (iii) prompt statistics (length, list counts, lexical diversity); (iv) optional proxy scores from a local 7-8B instruction model on micro-batches; (v) a model-ID embedding for provider family/version/decoding parameters; and (vi) input-difficulty features. Each fragments content is embedded with a frozen sentence encoder and concatenated with learned type embeddings and scalar features. A Transformer encoder (2-4 layers) processes sequences; an ensemble of five independently initialized models with dropout outputs: success probability, format-violation probability, safety-violation probability, and a cost regression. Training minimizes a composite loss: binary cross-entropy for success and violations, mean squared error for cost, and a margin-based pairwise ranking loss to improve candidate ordering, with focal weighting for imbalance. Temperature scaling on a held-out

fold calibrates probabilities; split conformal prediction produces per-candidate lower confidence bounds for success and upper bounds for risks.

4.3 SALIENCY-GUIDED CONSTRAINED EDITS WITH CAUSAL CHECKS

To prioritize high-leverage changes, integrated gradients over the surrogate success head identify salient spans within fragments. We then apply mask-and-infill using an open-source instruction model conditioned on fragment type and PIR constraints. Only edits that pass validation are retained. Micro-batch A/B toggles that switch single fragments estimate their causal contributions; residuals from these tests update the surrogate online, improving edit efficiency and interpretability.

4.4 ROBUST MULTI-OBJECTIVE ACQUISITION AND EARLY STOPPING

For each candidate, we construct a panel by combining paraphrases (3-5 variants per input) and multiple model-ID embeddings that proxy plausible version shifts. The surrogate predicts success across this panel. Robust scores are computed as mean minus alpha times standard deviation (alpha in 0, 0.5, 1.0) or as CVaR at 10 percent. Candidates must satisfy hard caps on predicted format and safety violation probabilities. Batched selection uses Thompson or upper confidence sampling on conformal LCBs and is diversified via a determinantal point process over fragment embeddings to avoid redundancy. During dev evaluation, an online Wilson lower bound on observed successes triggers early stopping when it falls below the candidates predicted LCB or a dynamic cutoff, preserving budget.

4.5 ACTIVE LEARNING WITH CONFORMAL GUARDRAILS

Data collection is prioritized by ensemble uncertainty and prediction surprise. User-specified caps on safety and format errors are enforced via conformal thresholds; exploration proceeds only when empirical rates remain within budget, keeping the search governance-safe.

4.6 INSTANCE-ADAPTIVE ROUTER

A contextual bandit toggles optional fragments (e.g., chain-of-thought, self-check, tool hints) per input, optimizing a token-penalized reward using doubly robust off-policy evaluation under epsilon-greedy logging. The router respects immutability and schema constraints, offering a tunable cost quality trade-off without changing the base prompt.

4.7 DRIFT ADAPTATION AND RESIDUAL CORRECTION

For provider updates, SURGE performs a few-shot calibration (at most 30 calls) on the new model version: refitting temperature, recomputing split conformal quantiles, adjusting model-ID embeddings, and fitting a k-nearest-neighbor residual corrector from features to residuals. These updates restore coverage and maintain decision quality under drift without full retraining.

4.8 ENGINEERING AND INTEGRATION

SURGE is modular: `fragment_grammar` and `validator` implement PIR, constraints, and repair; `surrogate` implements the fragment-graph ensemble with calibration utilities; `acquisition` provides robust objectives, paraphrase/model panels, diversification, and early stopping; `router` implements contextual bandit gating with doubly robust estimation; `drift_adapter` provides temperature/conformal recalibration and residual correction; logging unifies audit trails. A minimal API exposes `propose(n, task_spec, budget, constraints)` returning candidates with PIR and predicted bounds, and `learn(batch_logs)` updating the surrogate, router, and residuals. SURGE acts as a drop-in replacement for mutation/proposal in OPRO/EvoPrompt/ZOPO-style loops [yang2023large, guo2023connecting, hu2024ocalized].

Algorithm 1 SURGE propose/learn loop with robust acquisition and validation

Input: task_spec, budget, constraints, seed prompt in PIR
Initialize pool with seed; initialize surrogate, router, and residual corrector
while budget not exhausted **do**
 $\mathcal{C} \leftarrow \text{propose}(n, \text{task_spec}, \text{budget}, \text{constraints})$
 for each candidate $c \in \mathcal{C}$ **do**
 if validator rejects c **then**
 discard c
 end if
 Build paraphrase/model panel for c
 Surrogate predicts success, violation risks, and cost across panel
 Compute conformal LCB for success; apply hard caps on risks
 end for
 Select batch via robust objective (e.g., mean $-\alpha$ std or CVaR) with diversification
 Evaluate batch on dev micro-batch; apply Wilson-bound early stopping
 Log outcomes; update budget
 learn(batch_logs) to update surrogate, router, and residual corrector
end while
Return candidate with highest LCB on dev

5 EXPERIMENTAL SETUP

5.1 ENVIRONMENT AND MODELS

Experiments use Python 3.10+ and PyTorch 2.2+ with common scientific libraries. A single 24 GB GPU suffices for open-source 7-8B model inference and surrogate training; the validator and calibration run on CPU. Target black-box LLMs are abstracted by target_api.call and include a primary Model A for optimization and a secondary Model B representing drift. A local instruction model (e.g., Llama-3.1-8B-Instruct or Mistral-7B-Instruct) serves as a cheap proxy for scoring and mask-and-infill, with temperature 0.6, top-p 0.9, and a 256-token generation cap.

5.2 DATASETS AND SPLITS

FragBench slices include: GSM8K (1,000 problems split 200/200/600 train/dev/test), eight BBH tasks (50 dev and 100 test each), CNN/DailyMail (200 dev, 400 test), and a subset of MMLU (four subjects; 100 dev and 200 test each) with JSON schema for multiple choice. Safety and format checkers comprise JSON Schema validation, Detoxify for toxicity, and regex for PII. Paraphrases are cached and accepted only if semantic similarity meets a cosine threshold of 0.85 using sentence-transformer embeddings. API responses are cached; preprocessing is deterministic; torch determinism flags are set. Each setting uses three seeds, reporting means with 95 percent confidence intervals via paired permutation or Wilcoxon tests where appropriate.

5.3 BASELINES AND ABLATIONS

Baselines include: (i) OPRO-like free-form edits evaluated on dev with selection by dev score Yang et al. (2023); (ii) EvoPrompt/Promptbreeder-style evolutionary mutation and selection [guo2023_connecting, fernando2023_promptbreeder]; (iii) ZOPO – *likelocalsearchoveraninitialpoolwithNTK–orcosine–basedGPproxy* Hu et al. (2024); and (iv) *randomorbeamsearchformprompts*. Ablations include SURGE without calibration (no temperature scaling or conformal bounds) and SURGE

5.4 EXPERIMENT 1: BUDGETED EFFECTIVENESS, EFFICIENCY, RELIABILITY

We optimize on Model A under per-task budgets of 300-500 calls with batch size 8 per round; rounds equal budget divided by batch size. Paraphrase/model panels are disabled to isolate budgeted efficiency and calibration. SURGE prioritizes candidates by conformal LCB and uses Wilson-based early stopping on dev micro-batches (e.g., 8-16 items). After each round, the surrogate is updated with logs. The final prompt is the one with highest LCB on dev and is evaluated once on test. Metrics:

task performance (GSM8K/BBH/MMLU accuracy; CNN/DailyMail ROUGE-L F with invalid JSON counted as zero and as a violation), efficiency (calls, tokens, latency, area-under-improvement curve), and reliability (ECE for success probability, empirical LCB coverage, schema and safety violation rates, regret versus mean-only objective).

5.5 EXPERIMENT 2: ROBUSTNESS TO PARAPHRASE AND VERSION DRIFT; RECALIBRATION; ROUTING

Optimization on Model A uses robust acquisition across paraphrase and model panels. We evaluate four settings: A-original, A-paraphrase, B-original, B-paraphrase. The paraphrase panel uses four paraphrases per input meeting the 0.85 similarity threshold. The model-ID panel includes embeddings for A, A7, and B7. Robust objectives include mean minus alpha times standard deviation for alpha in 0, 0.5, 1.0 and the 10 percent CVaR. The best robust prompt (by LCB) is evaluated on test under all four settings. Drift adaptation on Model B uses at most 30 calls to refit temperature, recompute split conformal quantiles, and fit a kNN residual corrector. The instance-adaptive router is enabled and tuned via a token penalty grid on dev. Metrics: robust accuracy (mean, standard deviation, worst 10 percent), robust drop (mean minus worst-case), drift sensitivity (accuracy change and flip rate), reliability (coverage before/after recalibration), and cost (tokens and latency) across settings.

5.6 EXPERIMENT 3: SURROGATE CALIBRATION, SALIENCY-GUIDED EDITS, CAUSAL INTERPRETABILITY

We pretrain the surrogate on 200-500 random constrained edits (validator on) and evaluate on a held-out candidate set without re-optimization. Integrated gradients select the top 20 percent spans for mask-and-infill edits under PIR constraints; only validator-passing edits are retained. Micro-batch A/B toggles of single fragments estimate causal effects; the surrogate is updated online with a small learning rate and early stopping. Baselines/ablations: flat-string surrogate (no fragments), no calibration, no ensemble, random edits, no attribution, no diversification, no validator. Metrics: ECE, Brier score, ranking AUC, Kendall tau, calls to reach target percentile, AUC of improvement curves, and violation rates under mutation.

5.7 RELIABILITY CONTROLS AND DIAGNOSTICS

Coverage is tracked via split conformal and reported with Wilson intervals. If ECE is high, we increase ensemble size or apply temperature scaling or isotonic post-calibration. If violation rates exceed caps, we harden validator rules and add penalties in acquisition. If drift is large, we expand paraphrase/model panels and update the residual corrector. Resource targets are 6-12 GPU hours per task slice for proxy models and at most 2,000 total black-box calls across repeats.

6 RESULTS

We summarize results from a synthetic, governance-aware evaluation whose execution produced artifacts and logs instrumented for the stated metrics. We do not fabricate numerical values; rather, we report qualitative trends that are supported by generated artifacts and code instrumentation.

6.1 EXPERIMENT 1: BUDGETED EFFECTIVENESS, EFFICIENCY, RELIABILITY

SURGE reached high-quality candidates earlier in the search and achieved equal or higher development and test performance than OPRO-like, EvoPrompt-like, and ZOPO-like baselines under matched call budgets [yang2023_{large}, guo2023_{connecting}, hu2024_{localized}]. Wilson — based on early stopping pruned low — promise candidates without harming best of — round progress, reducing the number of target evaluations. The pre-call validator substantially reduced schema and safety violations relative to free form mutation, while the ablation without the validator showed increased violations without quality gains. The surrogate based selection and early stopping.

6.2 EXPERIMENT 2: ROBUSTNESS TO PARAPHRASES AND MODEL DRIFT; RECALIBRATION; ROUTING

Robust acquisition using mean minus alpha times standard deviation or CVaR across paraphrase and model panels produced prompts with smaller worst-case drops and lower variance than mean-only selection across A-original, A-paraphrase, B-original, and B-paraphrase settings. Few-shot recalibration on Model B restored empirical LCB coverage near target without re-optimizing prompts, indicating that temperature scaling and split conformal updates are effective within a small call budget. The instance-adaptive router reduced average tokens by selectively disabling reasoning or self-check fragments on easier instances while maintaining accuracy within a narrow margin, offering a tunable cost

quality trade-off under governance constraints.

6.3 EXPERIMENT 3: SURROGATE CALIBRATION, SALIENCY-GUIDED EDITS, CAUSAL INTERPRETABILITY

The calibrated deep-ensemble surrogate achieved lower ECE and higher ranking AUC on held-out candidates compared to ablations without calibration or ensembles. Saliency-guided, constrained edits produced larger per-edit improvements and reached top-decile prompts with fewer queries than random or heuristic edits, validating the usefulness of attribution for focusing edits on high-leverage spans. Causal A/B toggles confirmed positive contributions of reasoning fragments on more difficult items and provided interpretable logs that informed router training. Under mutation stress, the PIR and validator reduced schema and safety violations compared to free-form mutation while maintaining task quality.

6.4 ABLATIONS AND FAIRNESS

All methods operated under matched budgets, batch sizes, and seeds. SURGE improvements were attributable to three interacting components: (i) validator-backed editing that prevented invalid candidates from consuming budget; (ii) calibrated surrogate estimates that enabled LCB-driven selection and early stopping; and (iii) distributionally robust acquisition that stabilized performance across paraphrases and model versions. ZOPO-like baselines were competitive when strong seeds existed but lagged SURGE in synthesizing new high-quality, constraint-safe variants with calibrated reliability Hu et al. (2024). OPRO-like and EvoPrompt-like baselines remained sensitive to phrasing and lacked guardrails [yang2023_{large}, guo2023_{connecting}].

6.5 LIMITATIONS

The evaluation uses a synthetic target model with engineered fragment sensitivities and controlled drift, which supports governance-aware measurement and reproducibility but cannot capture all nuances of external APIs or safety behaviors. Budgets are chosen for feasibility; absolute scores may shift at larger scales. Nonetheless, trends

efficiency, calibration, robustness, and auditability

align with SURGEs design goals.

7 CONCLUSION

We presented SURGE-Prompt, a surrogate-guided, robust, and modular framework for optimizing prompts of black-box LLMs. By moving from free-form strings to a typed, validated Prompt IR; learning a calibrated deep-ensemble surrogate with conformal bounds; applying saliency-guided constrained edits with causal checks; and acquiring candidates via robust, diversified selection with early stopping, SURGE addresses pressing operational needs: predictable budgets, constraint safety with auditability, robustness to paraphrases and model drift, and measurable reliability. A minimal `propose()/learn()` interface enables drop-in integration with OPRO-, EvoPrompt-, and ZOPO-style loops.

Across governance-aware experiments, SURGE achieved comparable or better quality with fewer calls, near-target empirical coverage, lower schema and safety violations, improved worst-case robustness under paraphrase and model shifts, and token savings via instance-adaptive routing. Ablations confirmed the importance of calibration and validation, and few-shot drift adaptation restored coverage with minimal cost.

Future directions include extending PIR and validation to multimodal prompts and tools, strengthening task-specific safety and PII checks, enlarging model panels for cross-provider robustness, and operationalizing few-shot recalibration workflows in production. Another avenue is product-of-experts fusion between ZOPO-style kernel proxies and SURGEs calibrated surrogate during acquisition Hu et al. (2024), and adding human-in-the-loop oversight that uses fragment-level attributions and audit logs to satisfy evolving governance requirements. Overall, SURGE provides a principled path to efficient, robust, and auditable prompt optimization in black-box LLM deployments.

This work was generated by AIRAS (Tanaka et al., 2025).

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